INCH-POUND

MIL-E-82778(OS)
29 December 1989
SUPERSEDING
WS 22157B
9 December 1986
(See 6.5)

MILITARY SPECIFICATION

EXIT CONE ASSEMBLY, IN-PROCESS REQUIREMENTS FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the in-process requirements for one type of exit cone assembly, referred to herein as the assembly. The in-process requirements cover exit cone assembly characteristics which must be verified prior to assembly and fabrication of the end product (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-I-735 Isopropyl Alcohol

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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STANDARDS

MILITARY

MIL-STD-453 Inspection, Radiographic

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia Absorbent Gauze

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes

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precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **In-process.** The assembly shall conform to the requirements specified herein.

3.2 **Surface conditions.** The surface condition defects of the assembly shall not exceed the limits specified in table I after final machining.

TABLE I. Surface defect limits.

Component	Defect	Permissible Condition Extent
Retention ring, aft exit ring, aft liner, insert insulation and overwrap	Blemishes (see 6.3.1)	Shall not exceed 0.02 inch in depth. No two shall be closer together than 1 inch.
	Delaminations (see 6.3.2)	None allowed.
	Cracks (see 6.3.3)	None allowed.
Retention ring, aft exit ring, aft liner and overwrap	Wrinkled areas (see 6.3.4)	None allowed
Insert insulation	Wrinkled areas (see 6.3.4)	Wrinkle height (h) to wrinkle length (l) ratio (h/l) not greater than 0.20. No more than three in each part. None closer together than 1/2 inch.
Threads, o-ring surface, and grooved exit cone insert	Blemishes	None allowed.
Other coated surfaces of exit cone shell	Blemishes	None deeper than .002 inch.

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3.3 Subsurface condition. Defects in the subsurface condition of the assembly shall not exceed the limits specified in table II.

TABLE II. Subsurface defect limits.

Defect	Permissible Condition Extent
1. Cracks (see 6.3.3)	None allowed.
2. Low density lines (see 6.3.6)	
a. Aft Liner, retention ring, aft exit ring, and insert insulation	None wider than 0.030 inch. No more than five in each part. None closer together than 1 inch. (Low density line between the same two fabric plies which extends over part circumference is considered a single line.)
b. Overwrap	
(1) From retention ring/aft liner interface to forward end of exit cone insulation	None wider than 0.030 inch. No more than five in each part. None closer together than 1 inch. (Low density line between the same two fabric plies which extends over part circumference is considered a single line.)
(2) From retention ring/aft liner interface to aft end of exit cone insulation	None wider than 0.040 inch. No more than eight in part. No more than three in any 1-inch length. (Low density line between the same two fabric plies which extends over part circumference is considered a single line).
3. Low density areas (see 6.3.7)	Any single area less than 0.050 inch in length and 0.050 inch in width. No more than one in any 1-square-inch area. No more than five per part.
4. High density inclusion (see 6.3.8)	Any single inclusion shall be less than 0.040 inch in length or width in the plane of the radiograph. Total surface area of inclusions shall be less than 0.0016 square inch in any square inch.

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3.4 **Interface bonds.** Separations and voids in the interface bonds shall not exceed the limits specified in table III (see figure 1).

TABLE III. Interface bond criteria.

Permissible Extent (Unbonds)	
As Measured by Real Time Radiography (RTR)	As Measured on a Single Tangential Radiograph
1. Aft exit ring to retention ring. Accumulative unbonds or voided areas less than 5 percent of the total bond surface area (1/).	Accumulative unbonds or voided areas less than 25 percent of total bond line length.
2. Retention ring to aft liner. Accumulative unbonds or voided areas less than 5 percent of the total bond surface area (2/).	Accumulative unbonds or voided areas less than 25 percent of the total bond line length.
3. Overwrap to: a. Insert insulation b. Retention ring c. Aft exit ring d. Aft liner No single separation or void greater than 0.100 inch length by 0.050 inch wide by 10 degrees circumferentially with the exception of the step interfaces when a void or separation may extend 360 degrees circumferentially and be no greater than 0.050 inch long by 0.050 inch wide. No more than three in each part. None closer together than 1 inch.	No single separation or void greater than 0.100 inch length by 0.050 inch wide by 10 degrees circumferentially with the exception of the step interfaces when a void or separation may extend 360 degrees circumferentially and be no greater than 0.050 inch long by 0.050 inch wide. No more than three in each part. None closer together than 1 inch.
4. (M103) Exit cone insulator to exit cone shell. No single separations or void length greater than 0.50 inch. Accumulative separation or voids less than 10 percent of the bond length, measured longitudinally or circumferentially.	No single separations or void length greater than 0.50 inch. Accumulative separation or voids less than 10 percent of the bond length.

See footnotes at end of table.

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TABLE III. Interface bond criteria - Continued.

	Permissible Extent (Unbonds)
5. Grooved exit cone insert to insert insulation. Accumulative unbonds or voided areas less than 25 percent of the total bond line length, measured longitudinally or circumferentially.	Accumulative unbonds or voided areas less than 25 percent of the total bond line length.
6. Carbon cloth phenolic sleeve to ATJS graphite. Accumulative unbonds or voided areas less than 25 percent of the total bond line length, measured longitudinally or circumferentially.	Accumulative unbonds or voided areas less than 25 percent of the total bond line length.
7. Carbon cloth phenolic sleeve to insert insulation. Accumulative unbonds or voided areas less than 5 percent of the total bond surface area (3/).	Accumulative unbonds or voided areas less than 25 percent of the total bond line length.

1/ Percent unbond = $\dfrac{3.14\,[(\text{outer radius unbond})^2 - (\text{inner radius unbond})^2] \times (\text{unbond circumference length}) \times 100}{(10.9 \times 13.5)}$

Where: 10.9 in^2 = Total bond area
13.5 in = Average circumferential length

2/ Percent unbond = $\dfrac{3.14\,[(\text{outer radius unbond})^2 - (\text{inner radius unbond})^2] \times (\text{unbond circumference length}) \times 100}{(10.5 \times 14.8)}$

Where: 10.5 in^2 = Total bond area
14.8 in = Average circumferential length

3/ Percent unbond = $\dfrac{3.14\,[(\text{outer radius unbond})^2 - (\text{inner radius unbond})^2] \times (\text{unbond circumference length}) \times 100}{(2.4 \times 14.6)}$

Where: 2.4 in^2 = Total bond area
14.6 in = Average circumferential length

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3.5 **Workmanship.** The assembly shall be free from surface contamination, grease, oil, foreign material, and other defects which could render the part unsuitable for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table IV.

TABLE IV. Quality conformance inspection.

Inspection	Requirement	Test Method
Visual examination	3.2 & 3.5	4.3.1
Radiographic inspection	3.3 & 3.4	4.3.2

4.2.1 **Classification of characteristics.** The characteristics verified by inspections herein are classified as Major or Minor in accordance with WR-43. Inspections that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the inspection. Inspections not annotated with a classification code and serial number are classified Minor. When the classification does not apply to the complete paragraph, the portion to which the classification is applicable is enclosed in brackets (see 4.3.2).

4.3 **Methods of inspection.** The following inspection methods shall be used.

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4.3.1 Visual examination. The threads, o-ring surface and grooved exit cone insert of each assembly shall be visually examined to determine conformance to 3.2. Each assembly shall be visually examined to determine conformance to 3.2 and 3.5. Noncompliance with any specified requirement shall constitute cause for rejection of the assembly. The following solvent wipe method shall be used as an examination aid in determining conformance to 3.2.

a. Wet surface with methyl ethyl ketone (MEK) conforming to ASTM D 740 or isopropyl alcohol conforming to TT-I-735.

b. Wipe off excess solvent with type VIII absorbent gauze conforming to The United States Pharmacopeia.

c. Allow the part to dry in air. Defect will appear as a well-defined wetted line or area when surrounding surface area appears dry.

4.3.2 Radiographic inspection. Each assembly shall be radiographically inspected to determine the subsurface and interface bond condition as specified in 3.3 and 3.4. Noncompliance with any specified requirements shall be cause for rejection of the assembly. The inspection shall be subject to the following conditions:

a. Film radiography shall be in accordance with MIL-STD-453 and figure 2 herein.

b. Real time radiography (RTR) shall be in accordance with the contractor's written procedures. The procedures shall include but not be limited to the following:

(1) System description and operation.

(2) Process control, to include tests and inspections verifying video monitor resolution, grey scale range, electronic or geometrical magnification or both, and flatness of the as-viewed and recorded image.

(3) A RTR test procedure which is capable of consistently producing the desired inspection quality level. The test procedure shall include but not be limited to the following:

(a) A drawing or sketch of the component showing the location of the imaging device and penetrameter with respect to the radiation source.

(b) The kilovoltage, milliamperage and focal spot size of the radiation source.

(c) Radiation source to imaging device distance.

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(d) Type of video recording tape.

(e) Video field of view size, camera f stop, part rotation speed and axial index increment if applicable, and positive or negative video mode.

(f) Pertinent image processing parameters such as frame averaging, kernel size and shape, and video mixing ratio.

(g) When specified (see 6.2), a means of monitoring and verifying radiation dose rate.

c. The permanent record of inspection (x-ray film or video tape) shall demonstrate a steel wire penetrameter sensitivity of 1.5 percent.

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The exit cone assembly specified herein is intended for use as a component in the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. When a means of monitoring and verifying radiation dose rate must be included in the RTR test procedure (see 4.3.2b(3)(g)).

6.3 Definitions.

6.3.1 Blemishes. Any superficial imperfection or marks or deformities such as scratches, crazing, gouges, cuts, or stains.

6.3.2 Delamination. A separation between fabric plies which has sharply defined boundaries that are parallel to the fabric plies.

6.3.3 Crack. A parting or separation which extends across the fabric plies.

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6.3.4 **Wrinkled areas.** Distortions in fabric orientation normal to the plane of the fabric.

6.3.5 **Unbond.** A separation or void which occurs in the bond line between two components.

6.3.6 **Low density line.** A low density area between fabric plies which is parallel to the fabric plies.

6.3.7 **Low density areas.** A low density area which has appreciable width as measured perpendicular to the plies and is not necessarily oriented in any specific manner with respect to the plies.

6.3.8 **High density inclusion.** Object that has higher density than that of the surrounding material.

6.4 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.5 **Supersedure information.** MIL-E-82778 incorporates the following engineering change proposals (ECPs), notices of revision (NORs), and specification change notices (SCNs):

ECPs	NORs	SCNs
SSS1323 (1/27/86)	----	SCN 1 (7/8/87)
ARC001 (7/27/88)	ARC001.9 (7/27/88)	
ARC007 (9/29/88)	ARC007.1 (9/29/88)	----

Preparing Activity:
NAVY-OS
(Project 1340-N787)

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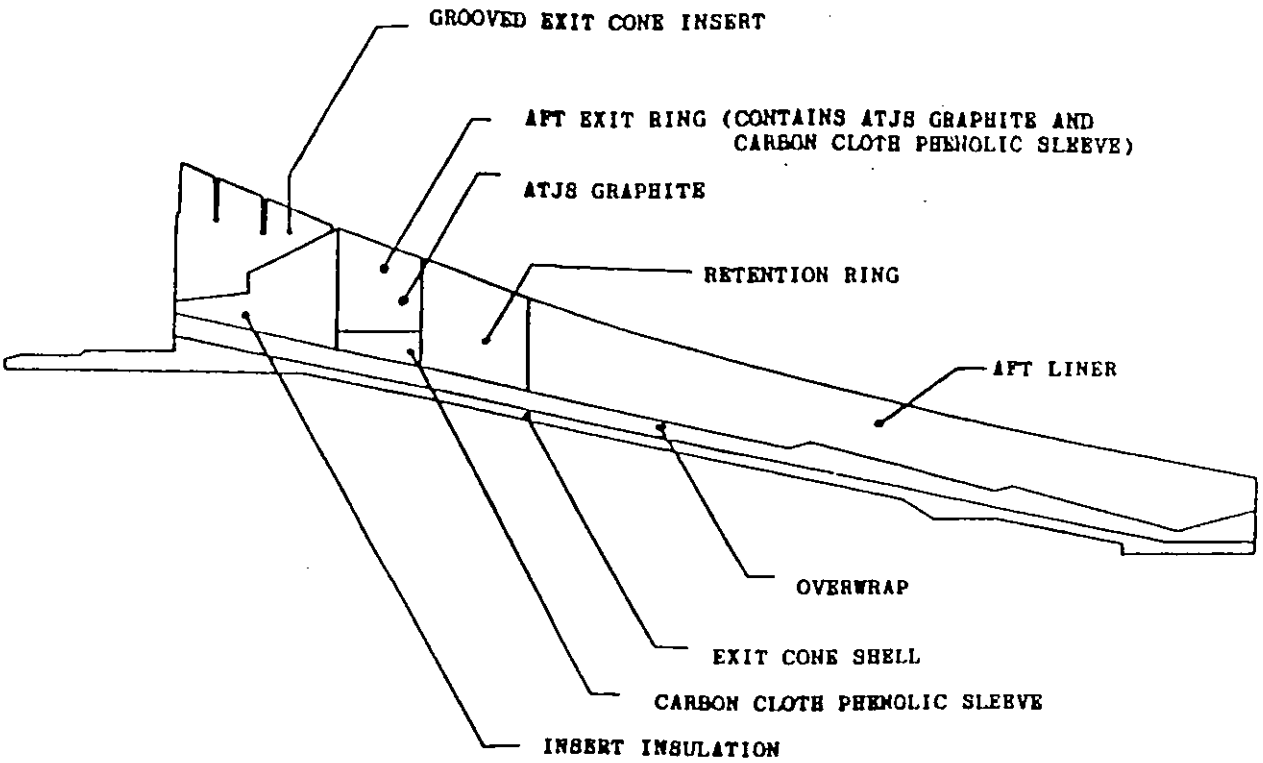
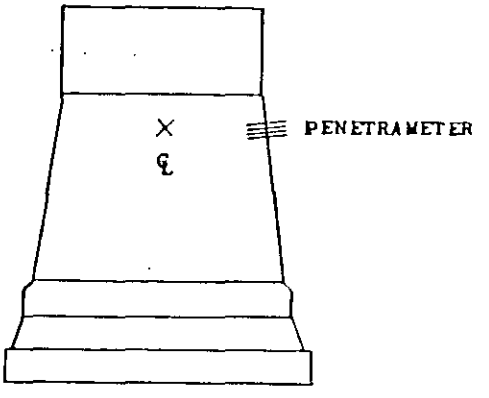


FIGURE 1. Exit cone assembly.

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NOTES

1. DEGREES SHALL BE MEASURED AFT LOOKING FORWARD IN A CLOCKWISE DIRECTION TO PRODUCE TANGENTIALS EVERY 60°.
2. IDENTIFY FILM, USING LEAD NUMBERS, WITH DEGREES AT TANGENT POINT AND LAST THREE DIGITS OF THE PART SERIAL NUMBER.
3. BEAM CENTERLINE SHALL BE LOCATED 3.00"+/-.100" AFT OF FORWARD END OF PART.



VIEW 1

FIGURE 2. X-ray technique.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-82778	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
EXIT CONE ASSEMBLY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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